ReliableML Method Note: Global and Local Explanations

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Abstract

The tabular explainer compares a Monte Carlo permutation estimate of interventional Shapley contributions with local weighted linear surrogate weights in the style of LIME. Global summaries aggregate absolute contributions over selected rows; local reports describe individual model outputs.

1 Inputs and outputs

The fitted estimator must provide `predict`, `predict_proba`, and `classes_`. Explanations require finite numeric features, at least two columns, and reference/background data. Binary explanations default to the positive class; multiclass use requires a selected class. `X` and background data must use the same feature schema.

The global summary is mean absolute Shapley value by feature over explained rows. Local results include signed Shapley contributions and LIME-style local surrogate weights. Reason alignment compares these methods using rank correlation, top-feature overlap, and sign agreement; bootstrap intervals can summarize rank agreement. An optional user-supplied global importance mapping is clearly distinguished from computed SHAP values. Optional exports include input columns, predictions, labels when supplied, and per-row top contributing features in CSV or supported serialized formats.

2 Methods

The Shapley approximation samples feature permutations and compares model probabilities along each permutation against a reference background. Its Monte Carlo error depends on sample count and feature dependence. The local surrogate perturbs observations, weights nearby samples, and fits a local linear explanation; fidelity depends on the perturbation and neighborhood design. These are model-behavior explanations, not tests of causal effects.

3 Interpretation and limits

Agreement between attribution methods indicates explanation consistency under their respective approximations. It does not demonstrate that a prediction is correct, that features are valid reasons, or that no spurious correlation is present. Explanations can be sensitive to background choice, correlated features, perturbation realism, and model probability quality. Use representative labeled holdout evaluation, subgroup/time analysis, leakage checks, calibration, realistic counterfactuals, and domain review. Causal claims require a defensible causal model and assumptions.

4 Interface

```
report = model.explain(X_test, class_label=1)
print(report.global_shap_importance)
print(report.local_explanations[0])
print(report.reason_alignment)
```

An already-fitted external estimator can be passed to `explain_estimator`; labels are optional for explanations and required only for the optional descriptive label evaluation.

5 Further reading

References

[1] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 2017. https://arxiv.org/abs/1705.07874.

[2] M. T. Ribeiro, S. Singh, and C. Guestrin. "Why should I trust you?" Explaining the predictions of any classifier. *Proceedings of KDD*, 2016. https://doi.org/10.1145/2939672.2939778.

[3] ReliableML explanation implementation: https://github.com/Jabualani/ML-Reliabilty.